10TH CLASS MATHEMATICS — NEW SYLLABUS (PECTAA)

Review Exercise 7 — Trigonometry

Unit 7 | Review + MCQs | New Book (PECTAA)

Review Exercise 7 (Solved)

The review covers the whole unit: signs and values of ratios, the law of sines, the law of cosines, area of a triangle, three-dimensional problems and three-figure bearings.

1. Four possible answers are given for the following questions. Choose the correct answer.

(i) What is the value of cot 60°?

(a) 0 (b) 1
(c) √3 (d) 1/√3

Answer: (d) 1/√3 — cot 60° = 1 ÷ tan 60° = 1/√3

(ii) All trigonometric ratios are positive in:

(a) 1st quadrant (b) 2nd quadrant
(c) 3rd quadrant (d) 4th quadrant

Answer: (a) 1st quadrant — in the first quadrant x and y are both positive

(iii) cosec θ is positive in:

(a) I & III quadrants (b) II & IV quadrants
(c) I & II quadrants (d) I & IV quadrants

Answer: (c) I & II quadrants — cosec θ has the sign of sin θ, which is positive in I and II

(iv) sin(90° + θ) =

(a) sin θ (b) − sin θ
(c) cos θ (d) − cos θ

Answer: (c) cos θ — sin(90° + θ) lies in the second quadrant and equals cos θ

(v) tan(180° − θ) =

(a) tan θ (b) cot θ
(c) − cot θ (d) − tan θ

Answer: (d) − tan θ — second quadrant, tan is negative and the ratio stays tan

(vi) The law of sines is:

(a) b/sin α = a/sin β = c/sin γ (b) a/sin α = c/sin β = b/sin γ
(c) a/sin α = b/sin β = c/sin γ (d) a/sin α = a/sin β = a/sin γ

Answer: (c) a/sin α = b/sin β = c/sin γ — each side is paired with the sine of its opposite angle

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(vii) The law of cosines is:

(a) $\cos\alpha = (b^2 + a^2 - c^2)/2ab$ (b) $\cos\beta = (c^2 + b^2 - a^2)/2cb$

(c) $\cos\gamma = (a^2 + b^2 - c^2)/2ab$ (d) $\cos\gamma = (ca^2 + b^2 - c^2)/2ab$

Answer: (c) $\cos\gamma = (a^2 + b^2 - c^2)/2ab$ — cos γ uses the two sides a, b that contain γ

(viii) Area of △ABC =

(a) ½ ab sin α (b) ½ bc sin β

(c) ½ sin γ (d) ½ ac sin β

Answer: (d) ½ ac sin β — the two sides must be the ones containing the angle

(ix) Bearing is measured from:

(a) East (b) West

(c) North (d) South

Answer: (c) North — a bearing always starts at the North line

(x) Bearing is written as a:

(a) 1 figure (b) 2 figures

(c) 3 figures (d) 4 figures

Answer: (c) 3 figures — for example 015°, 140°, 300°

2. Calculate the area of △ABC, in which

(i) a = 4 cm, b = 6 cm, c = 8 cm (ii) b = 2.1 cm, c = 5 cm, γ = 45°

(iii) c = 3.1 cm, γ = 44°, α = 36°

(i) a = 4 cm, b = 6 cm, c = 8 cm

Three sides are given, so use Heron's formula

$s = \frac{4 + 6 + 8}{2} = 9$

Area = $\sqrt{s(s - a)(s - b)(s - c)} = \sqrt{(9 \times 5 \times 3 \times 1)}$

$= \sqrt{135} = 11.62$ cm²

Answer: Area ≈ 11.62 cm²

(ii) b = 2.1 cm, c = 5 cm, γ = 45°

$\sin\beta = (b \sin\gamma) / c = (2.1 \times 0.7071) / 5 = 0.2970$

$\beta = 17.28°$

$\alpha = 180° - 45° - 17.28° = 117.72°$

Area = $\frac{1}{2}$ bc sin α = $\frac{1}{2} \times 2.1 \times 5 \times \sin 117.72°$

$= 5.25 \times 0.8852 = 4.65$ cm²

Answer: Area ≈ 4.65 cm²

(iii) c = 3.1 cm, γ = 44°, α = 36°

$\beta = 180° - 44° - 36° = 100°$

$a = (c \sin\alpha) / \sin\gamma = (3.1 \times 0.5878) / 0.6947 = 2.62$ cm

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Area = $\frac{1}{2}$ ac sin β = $\frac{1}{2}$ × 2.62 × 3.1 × sin 100°
= 4.06 × 0.9848 = 4.00 cm²

Answer: Area ≈ 4.00 cm²

3. Solve the triangle ABC, in which

(i) a = 5.4 cm, b = 3.4 cm, α = 49° (ii) α = 32°, γ = 48°, c = 81 cm

(i) a = 5.4 cm, b = 3.4 cm, α = 49°

sin β = (b sin α) / a = (3.4 × 0.7547) / 5.4 = 0.4752
β = 28.4°
γ = 180° − 49° − 28.4° = 102.6°
c = (a sin γ) / sin α = (5.4 × 0.9759) / 0.7547 = 6.98 cm

Answer: β ≈ 28.4°, γ ≈ 102.6°, c ≈ 6.98 cm

(ii) α = 32°, γ = 48°, c = 81 cm

β = 180° − 32° − 48° = 100°
a = (c sin α) / sin γ = (81 × 0.5299) / 0.7431 = 57.8 cm
b = (c sin β) / sin γ = (81 × 0.9848) / 0.7431 = 107.3 cm

Answer: β = 100°, a ≈ 57.8 cm, b ≈ 107.3 cm

4. The diagram shows a cuboid ABCDEFGH in which AB = 5 cm, BC = 7 cm and AE = 3 cm. (i) Calculate the length of AG (correct to 3 significant figures). (ii) Calculate the size of the angle between AG and the plane ABCD (correct to 1 decimal place).

AE is the height, AB and BC are the edges of the base, and AC is the diagonal of the base ABCD.

(i) length of AG

In the base, $AC^2 = AB^2 + BC^2 = 5^2 + 7^2 = 25 + 49 = 74$
In the right triangle ACG, $AG^2 = AC^2 + CG^2$
$AG^2 = 74 + 3^2 = 74 + 9 = 83$
AG = √83 = 9.11 cm

Answer: AG ≈ 9.11 cm

(ii) angle between AG and the plane ABCD

AC is the projection of AG on the base, so the required angle is ∠GAC
tan θ = $\frac{CG}{AC}$ = $\frac{3}{\sqrt{74}}$ = $\frac{3}{8.602}$ = 0.3488
θ = 19.2°

Answer: Angle between AG and ABCD ≈ 19.2°

5. The diagram shows a triangular prism ABCDEF in which AB = 8 cm, BF = 14 cm and EF = 19 cm. (i) Calculate the distance between A and F. (ii) Calculate the angle between AF and the plane BCEF.

AB is perpendicular to the rectangular face BCEF, so AB is perpendicular to BF and BF is the projection of AF on this face.

(i) distance between A and F

In the right-angled triangle ABF

$AF^2 = AB^2 + BF^2 = 8^2 + 14^2 = 64 + 196 = 260$

$AF = \sqrt{260} = 16.12$ cm

Answer: AF ≈ 16.12 cm

(ii) angle between AF and the plane BCEF

The projection of AF on the plane BCEF is BF

$\tan\theta = \frac{AB}{BF} = \frac{8}{14} = 0.5714$

$\theta = 29.7°$

Answer: Angle between AF and BCEF ≈ 29.7°

6. Hashim walks 4 km due North, then turns and walks 3 km due East. What is the bearing from the starting point to his final position?

$\tan\theta = \frac{3}{4} = 0.7500$

$\theta = 36.9°$ (measured from North towards East)

Answer: Bearing ≈ 037°

7. A pilot flies 200 km on a bearing of 045°, then turns and flies 150 km on a bearing of 135°. How far is the plane from its original position?

First leg: East = 200 sin 45° = 141.42 km, North = 200 cos 45° = 141.42 km

Second leg: East = 150 sin 135° = 106.07 km, North = 150 cos 135° = − 106.07 km

Total East = 141.42 + 106.07 = 247.49 km

Total North = 141.42 − 106.07 = 35.35 km

$d^2 = 247.49^2 + 35.35^2 = 61251 + 1250 = 62501$

$d = \sqrt{62501} = 250$ km

Answer: Distance from the original position = 250 km

Quick Revision — Unit 7 at a Glance

$\sin(180° - \theta) = \sin\theta$, $\cos(180° - \theta) = -\cos\theta$, $\tan(180° - \theta) = -\tan\theta$.

$\sin(90° + \theta) = \cos\theta$, $\cos(90° + \theta) = -\sin\theta$, $\tan(90° + \theta) = -\cot\theta$.

Law of sines: $a \div \sin\alpha = b \div \sin\beta = c \div \sin\gamma$.

Law of cosines: $a^2 = b^2 + c^2 - 2bc\cos\alpha$.

Area = ½ × (two sides) × sin (the angle between them); Heron's formula when three sides are given.

Bearing: measured clockwise from North, written in three figures; reverse bearing differs by 180°.